Week 7, Lecture 2: Bayesian Inference

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** So far we have treated a parameter as a fixed but unknown number and described our uncertainty with confidence intervals. Bayesian inference describes that uncertainty with a probability distribution over the parameter values. Today we build that idea from Bayes' rule, work one example all the way through, and talk about why Bayesian methods have become so common.
- **Read:** Bolker 6.2.2 (Bayesian analysis, printed p. 237-243) and 6.4.2 (posterior and marginal distributions, p. 258-260).
- **Review:** Bayes' rule from Week 3 (Bolker 4.3, p. 145-152, especially 4.3.3 on priors), and the Beta distribution (Bolker 4.5.2.5, p. 176). Look back at the Week 5 WOD (*Writing the Likelihood*) and your likelihood values from Part 2.

Overview (instructor only)

This file has two parts, both instructor-only. **Part 1** is read beforehand. **Part 2** (after the page break, large print) is the podium copy.

As taught (Wed., Oct. 7). The period covered:

- logistic regression and the log-odds (not planned; Lab 6 used a logistic dose-response model that hadn't come up in lecture), now written up as the opening section below;
- section 1 (big picture: parameter uncertainty as a probability distribution);
- section 2 (Bayes' rule applied to parameter values, with the focus on the numerator, prior $\times$ likelihood; the worked nest example);
- section 3 (the Beta-binomial conjugacy math);
- flat priors on different scales (a flat prior on p isn't flat on the log-odds), covered well enough that Week 8 Monday treats it as a callback only.

No discussion period. The rest of the original plan (the germination example, summarizing a posterior, where new Bayesians get stuck, why Bayesian methods are so widely used, and the discussion questions) moved to `NOTES_Week8_Lecture1_PriorsAndPosteriors.Rmd`, which starts with the germination example.

No WOD today (as taught, Wed., Oct. 7). The planned take-home WOD, *Curvature by Hand*, was not handed out because students are working on project propos-

als. It was later handed out as an optional take-home on Wed., Oct. 14, renamed to `wod/WOD_Week8_Lecture2_CurvatureByHand.Rmd` (key in `wod/answer_keys/`).

Source material: Bolker 6.2.2.1 (binomial/Beta conjugate example); legacy `LECTURE6.Rmd` ("Bayesian Analysis #1: Concepts": prior × likelihood figures, brute-force Bayes). The grid example reuses the Week 5 WOD nest data, because students already computed those likelihoods by hand.

Where this sits in the week

- **Monday:** uncertainty from the shape of the likelihood: curvature and the likelihood ratio interval (pikas: [1.82, 4.61]). The planned Bayesian preview didn't fit, so today is the first look at priors and posteriors.
- **Tuesday (Lab 6):** opened with the quadratic approximation, the Hessian, and slices vs. profiles (moved from Monday), then Bliss's beetle fumigation data: profile and Hessian intervals for the LD50.
- **Today:** Bayesian inference from the ground up. Keep the math to one-parameter examples; MCMC and Stan come later.
- **Coming up:** Week 8 Monday continues with the germination example and the rest of the Bayesian material, then starts optimization (Bolker Ch. 7) if time; MCMC and Lab 7 on Bayesian inference follow in Week 9.

Part 1: Lecture content

Opening: logistic regression and the log-odds (~10 min, added in class)

Lab 6 fit a logistic dose-response curve to the beetle data, but lecture hadn't covered the logistic function or the log-odds. Written up here as taught, with a neutral example (not the lab's numbers).

The model. For a probability that changes with a covariate x:

$$p(x) = \frac{e^{a+bx}}{1+e^{a+bx}} = \frac{1}{1+e^{-(a+bx)}}.$$

It's S-shaped and stays between 0 and 1 for any a, b, and x (Bolker's bestiary, Ch. 3).

The log-odds. The odds are $p/(1-p)$ (e.g. $p = 0.75$ is odds of 3, "3 to 1"). Taking logs of the odds undoes the logistic:

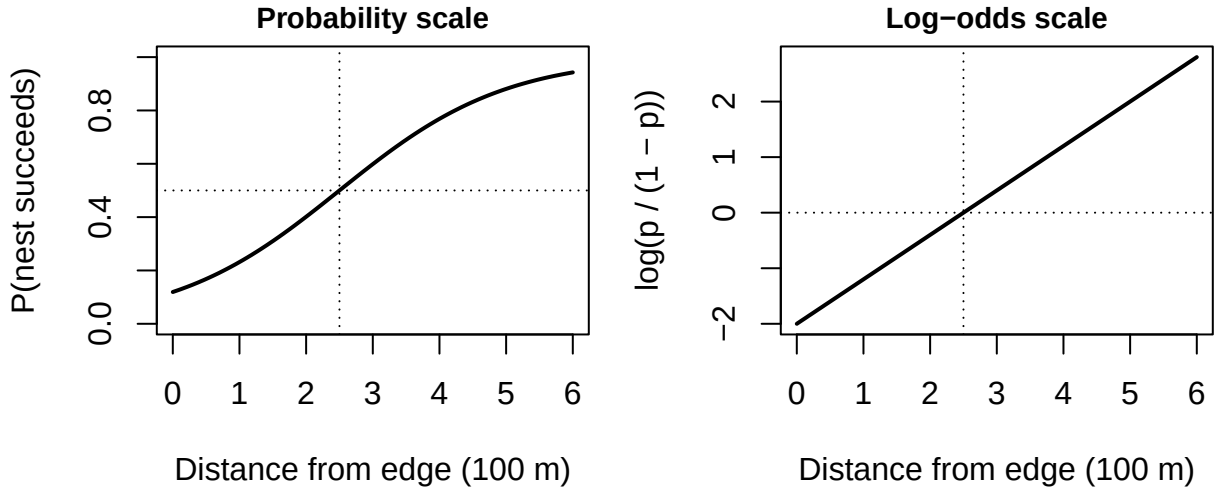
$$\log \frac{p}{1-p} = a + bx.$$

So logistic regression is a straight line on the log-odds scale.

- b is the change in log-odds per unit of x; e^b multiplies the odds.
- $p = 0.5$ where $a + bx = 0$, so at $x = -a/b$ (the LD50 in Lab 6).

- In R: `plogis()` is the logistic function, `qlogis()` the log-odds.

```
a_ex <- -2; b_ex <- 0.8   # nest success vs. distance from edge (100s of m)
x_seq <- seq(0, 6, length.out = 200)
par(mfrow = c(1, 2), mar = c(4, 4.5, 1.5, 0.5))
plot(x_seq, plogis(a_ex + b_ex * x_seq), type = "l", lwd = 2, ylim = c(0, 1),
     xlab = "Distance from edge (100 m)", ylab = "P(nest succeeds)",
     main = "Probability scale", cex.main = 0.9)
abline(h = 0.5, v = -a_ex / b_ex, lty = 3)
plot(x_seq, a_ex + b_ex * x_seq, type = "l", lwd = 2,
     xlab = "Distance from edge (100 m)", ylab = "log(p / (1 - p))",
     main = "Log-odds scale", cex.main = 0.9)
abline(h = 0, v = -a_ex / b_ex, lty = 3)
```



Sticking point: odds and probability get used interchangeably in casual speech. Odds of 3 is a probability of 0.75, not 3.

The log-odds scale comes back with priors: a prior that is flat on p is not flat on the log-odds (next session).

1. Big picture (~3 min)

Loop on the board:

$$\text{THEORY} \rightarrow \text{MODEL(S)} \rightarrow \text{DATA (fit)} \rightarrow \text{COMPARE} \rightarrow \text{repeat}$$

- The model and the likelihood don't change today. What changes is how we turn the likelihood into a statement about the parameter.
- **Frequentist view (Monday):** the parameter is a fixed, unknown number. The data are random. A 95% confidence interval is a procedure that captures the true value in 95% of repeated experiments.

- **Bayesian view (today):** our uncertainty about the parameter is itself described by a probability distribution over the parameter values. Before the data, that distribution is the **prior**; after, it's the **posterior**. The likelihood is what connects them.
- The tool that does the updating is one students already know: Bayes' rule, from Week 3 and Lab 3's camera-trap classifier.

2. Bayes' rule, from a few hypotheses to a whole distribution (~7 min)

Recall Week 3:

$$P(H_i \mid D) = \frac{P(D \mid H_i)\, P(H_i)}{\sum_j P(D \mid H_j)\, P(H_j)}.$$

In Lab 3, the hypotheses were "target species" and "not." Now make the hypotheses be **values of a parameter**.

Step 1: three hypotheses. Week 5 WOD nest data: 4 nests, outcomes 1, 1, 0, 1, so $\mathcal{L}_4(p) = p^3(1-p)$. Treat $p = 0.25, 0.5, 0.75$ as three competing hypotheses, equally likely beforehand.

```
p3 <- c(0.25, 0.5, 0.75)
lik3 <- p3^3 * (1 - p3)
prior3 <- rep(1/3, 3)
post3 <- lik3 * prior3 / sum(lik3 * prior3)
round(rbind(likelihood = lik3, posterior = post3), 3)
```

```
##             [,1]  [,2]  [,3]
## likelihood 0.012 0.062 0.105
## posterior  0.065 0.348 0.587
```

```
sum(lik3)          # Week 5 WOD 2d: likelihoods don't sum to 1
```

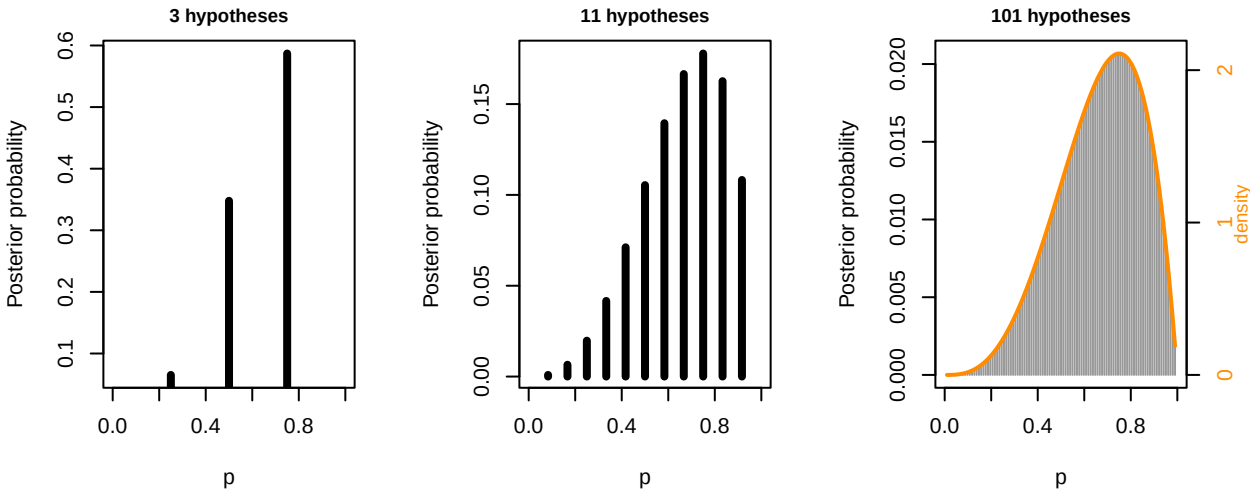
```
## [1] 0.1796875
```

- Students computed these likelihoods in the Week 5 WOD (3/256, 16/256, 27/256) and found they sum to 46/256, not 1. Bayes' rule fixes that: multiply by the prior, divide by the total. Posterior probabilities 3/46, 16/46, 27/46.
- With equal priors, the posterior is just the likelihood rescaled to sum to 1 (Bolker eq. 4.3.16).

Step 2: more hypotheses. Nothing stops us using 100 or 10,000 values of p, still with equal prior weight on each. Below, the same calculation with 3, 11, and 101 values:

```
# posterior probability at each of n_grid equally spaced values of p (flat prior)
grid_post <- function(n_grid) {
  p_g <- seq(0, 1, length.out = n_grid + 2)[-c(1, n_grid + 2)]
  w <- p_g^3 * (1 - p_g)
  list(p = p_g, prob = w / sum(w), spacing = diff(p_g)[1])
```

```
}
# bars = probabilities; on the finest grid, overlay Beta(4, 2) density x spacing
plot_grids <- function(ylab = "Posterior probability", titles = TRUE) {
  par(mfrow = c(1, 3), mar = c(4, 4, 2, 2.5))
  for (n_grid in c(3, 11, 101)) {
    g <- grid_post(n_grid)
    plot(g$p, g$prob, type = "h", lwd = if (n_grid < 20) 4 else 1,
         col = if (n_grid < 20) "black" else "grey60",
         xlim = c(0, 1), xlab = "p", ylab = ylab, cex.main = 0.9,
         main = if (titles) paste(n_grid, "hypotheses") else n_grid)
    if (n_grid == 101) {
      lines(g$p, dbeta(g$p, 4, 2) * g$spacing, lwd = 2, col = "darkorange")
      axis(4, at = (0:2) * g$spacing, labels = 0:2, col.axis = "darkorange")
      mtext("density", side = 4, line = 1.6, cex = 0.6, col = "darkorange")
    }
  }
}
plot_grids()
```




```
grid_tab <- t(sapply(c(3, 11, 101), function(n_grid) {
  g <- grid_post(n_grid)
  c(n = n_grid, tallest_bar = max(g$prob),
    bar_over_spacing = max(g$prob) / g$spacing)
}))
round(grid_tab, 3)
```

```
##        n tallest_bar bar_over_spacing
## [1,]   3       0.587            2.348
## [2,]  11       0.178            2.134
## [3,] 101       0.021            2.109
```

```r
dbeta(0.75, 4, 2)   # peak of the limiting density
```

```
## [1] 2.109375
```

- **The bars shrink.** The tallest bar goes from 0.587 to 0.178 to 0.021. With more hypotheses the same total probability is split among more of them, and with infinitely many, each single value of p has probability zero.
- **Divide by the spacing and it settles down.** Probability per unit of p (bar $\div$ spacing) gives 2.35, 2.13, 2.11: by 101 values it matches the peak of the orange curve (2.11). That curve is a probability **density**, Beta(4, 2) (section 3 explains the name). Probabilities now come from areas under it.
- So the sum becomes an integral and the prior and posterior become densities:

$$\underbrace{P(p \mid y)}_{\text{posterior}} = \frac{\overbrace{\mathcal{L}(p)}^{\text{likelihood}}\ \overbrace{P(p)}^{\text{prior}}}{\int_0^1 \mathcal{L}(p)\, P(p)\, dp}$$

- The denominator is a single number, the **marginal likelihood**: the probability of the data averaged over all parameter values. It only rescales the curve to area 1. For one parameter it's easy (a sum or an integral). For 20 parameters it's a 20-dimensional integral, which is why MCMC exists (later in the semester).
- Notation: the posterior $P(p \mid y)$ is a real conditional probability, so it gets a bar. The likelihood $\mathcal{L}(p)$ still doesn't.

Sticking point: students conflate "a probability distribution for p" with "p is random, like a coin flip." Say plainly: to a Bayesian the posterior describes *our knowledge* of p, not variation in p itself. The nest success rate is one number; we're uncertain about it.

Second sticking point: "the posterior probability that $p = 0.75$ is 2.1." A density isn't a probability (it can exceed 1). Ask the room why the bars keep shrinking before giving the answer.

3. The Beta prior and conjugacy (~8 min)

For k successes in n binomial trials, the likelihood is $\mathcal{L}(p) \propto p^k(1-p)^{n-k}$.

The prior. A Beta(α, β) density is $P(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$. Same shape as a binomial likelihood. Multiply them (the numerator of Bayes' rule):

$$P(p \mid y) \propto p^{k+\alpha-1}(1-p)^{n-k+\beta-1},$$

which is another Beta: **Beta($\alpha + k$, $\beta + n - k$)**. A prior that gives a posterior from the same family is called **conjugate**. It means no integral: just add counts.

- Interpretation (Bolker p. 237): $\alpha - 1$ and $\beta - 1$ act like earlier successes and failures. Beta(1, 1) is flat: no prior data.

- The nests: 3 of 4 with a flat prior gives Beta(4, 2), the orange curve in the grid figure above.

Sticking point: the proportionality sign. The constant in front of the Beta density gets dropped while multiplying and comes back at the end, because a density has to integrate to 1. That's the denominator of Bayes' rule doing its job.

4. Return to the big picture (~1 min)

- Same model, same likelihood. Bayesian inference multiplies by a prior and reads uncertainty off a posterior distribution.
- With one parameter and a conjugate prior, it's just adding counts. Next time: a worked example with two different priors.

PART 2: PODIUM CHEAT SHEET

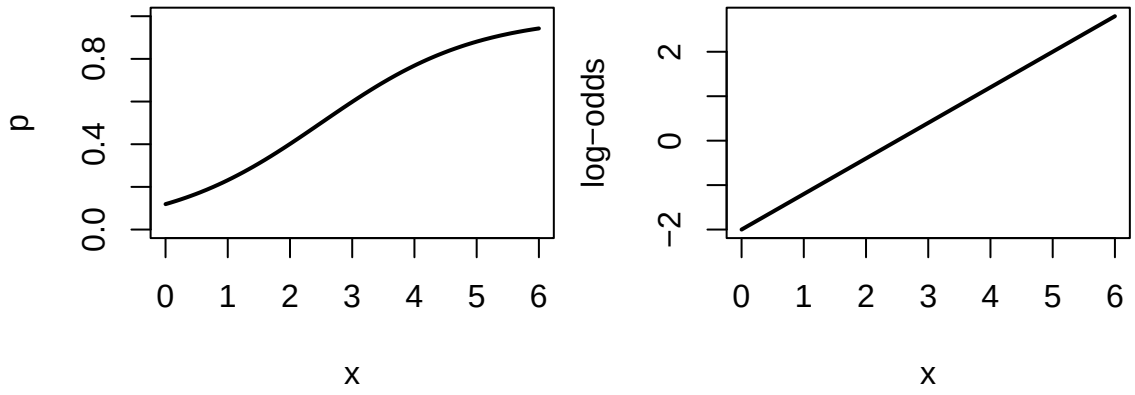
Terse on purpose. Full rationale and code: Part 1. As taught, Oct. 7.

No WOD. Logistic ~10 min. Lecture ~35 min.

Opening: logistic regression (~10 min)

$$p = \frac{1}{1 + e^{-(a+bx)}} \qquad \log \frac{p}{1-p} = a + bx$$

- Odds $= p/(1-p)$; $p = 0.75$ is odds 3
- Straight line on the log-odds scale; e^b multiplies the odds
- $p = 0.5$ at $x = -a/b$ (Lab 6 LD50)
- `plogis()`, `qlogis()`



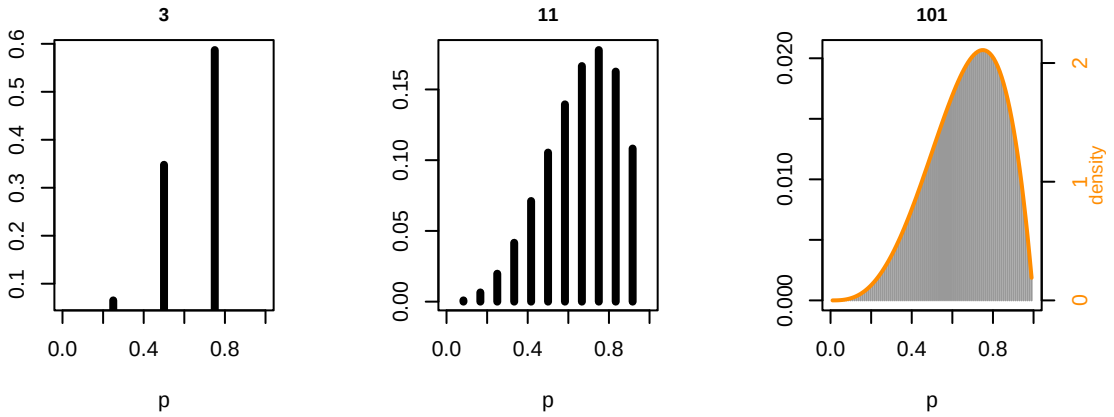
1. Big picture (~3 min)

- Loop on board: THEORY → MODEL → DATA (fit) → COMPARE → repeat
- Same likelihood; new way to describe uncertainty
- Frequentist: parameter fixed, data random, CI = procedure
- Bayesian: uncertainty about parameter = probability distribution
- Prior → (likelihood) → posterior, via Bayes' rule (Week 3, Lab 3)

2. Bayes' rule, few hypotheses to many (~7 min)

$$P(H_i \mid D) = \frac{P(D \mid H_i)P(H_i)}{\sum_j P(D \mid H_j)P(H_j)}$$

- Week 5 nests: $\mathcal{L}_4(p) = p^3(1-p)$
- $p = 0.25, 0.5, 0.75$: likelihoods 3, 16, 27 (/256), sum 46/256
- Equal priors → posterior 3/46, 16/46, 27/46



- Flat prior on every grid
- Ask: why do the bars shrink? (0.59, 0.18, 0.021)
- Bar ÷ spacing: 2.35, 2.13, 2.11 → density, Beta(4, 2) peak 2.11
- One value of p has probability 0; areas give probabilities

$$P(p \mid y) = \frac{\mathcal{L}(p)\, P(p)}{\int \mathcal{L}(p)\, P(p)\, dp}$$

- Denominator = one number; hard in many dimensions → MCMC
- **Sticking points:** posterior = our knowledge of p, not p varying; density ≠ probability (can exceed 1)

3. Beta prior and conjugacy (~8 min)

- Binomial: $\mathcal{L}(p) \propto p^k(1-p)^{n-k}$
- Beta prior: $P(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$
- Numerator: $p^{k+\alpha-1}(1-p)^{n-k+\beta-1} \to \text{Beta}(\alpha+k,\, \beta+n-k)$
- Conjugate = add counts; $\alpha-1$, $\beta-1$ = prior successes, failures
- Beta(1, 1) = flat; nests: Beta(4, 2)
- Constant dropped, comes back so area = 1

4. Wrap (~1 min)

- Next time: germination example, two priors